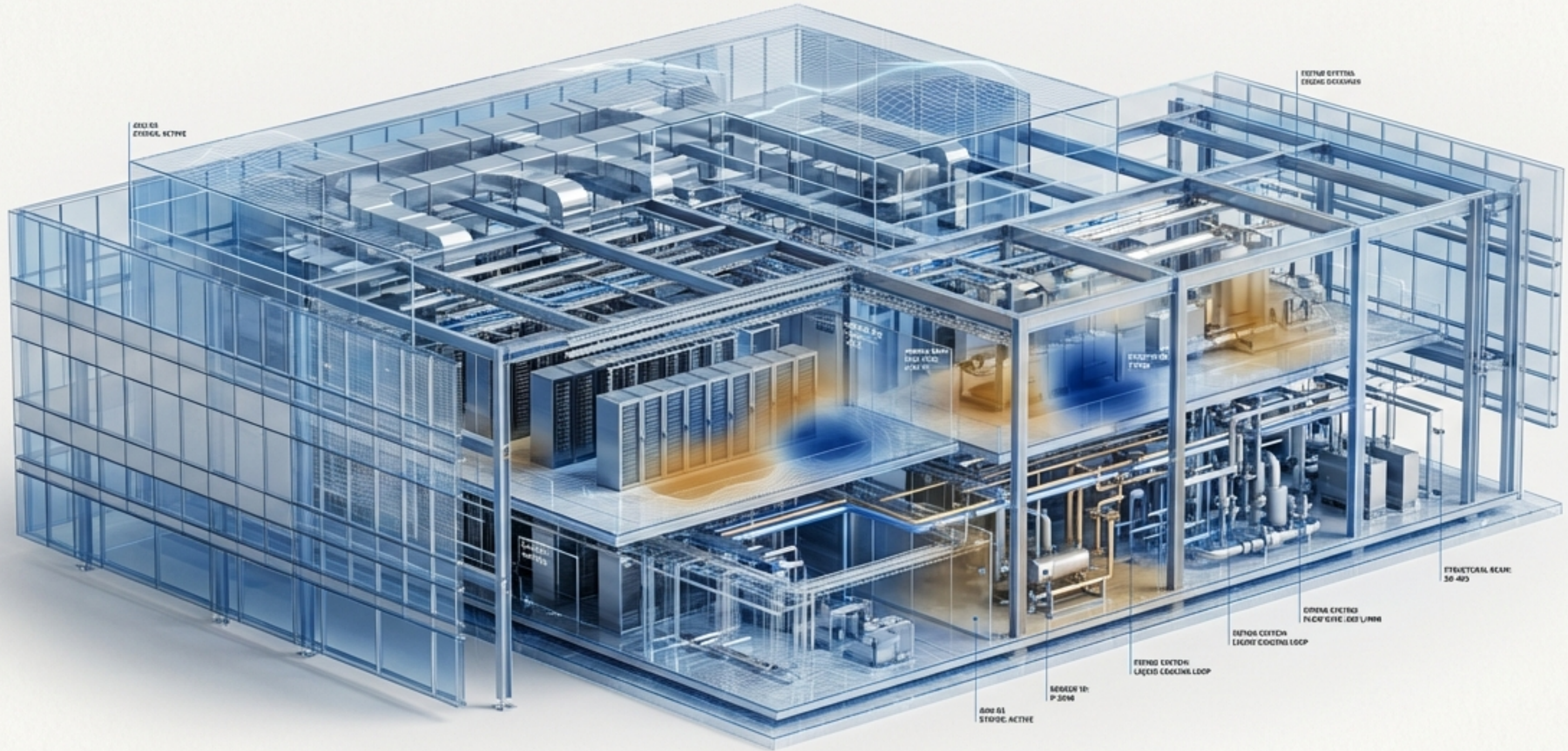


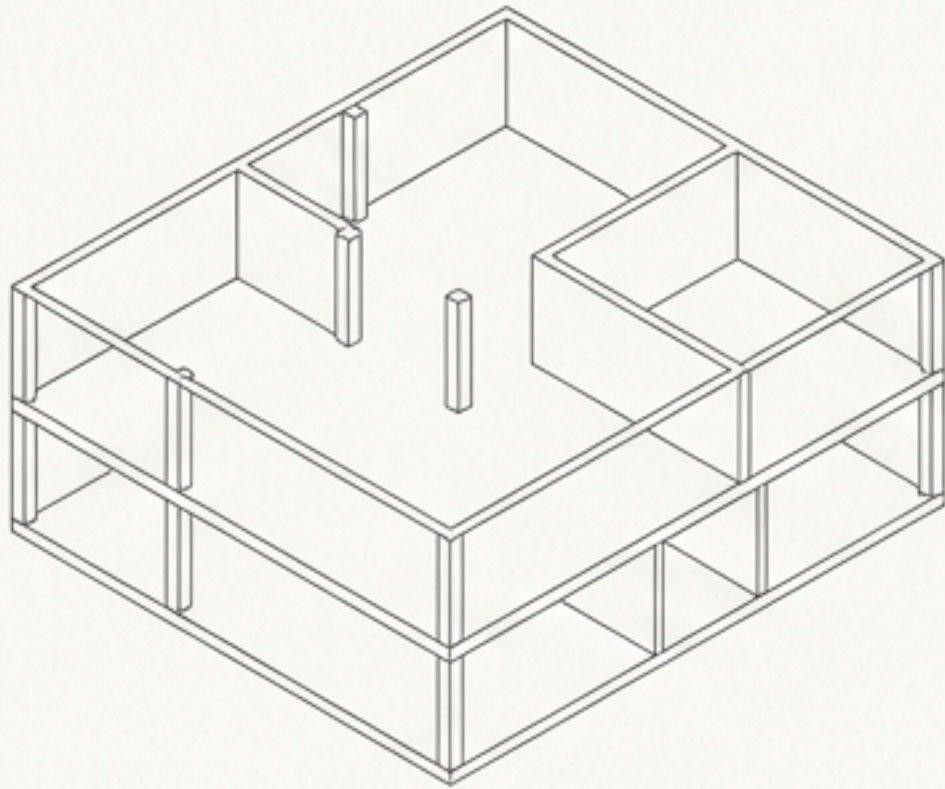
Agent-02: The Complete Space Ontology

From Architectural Blueprint to Operational Digital Twin



A Deliberate Evolution from a Static Plan to a Dynamic System

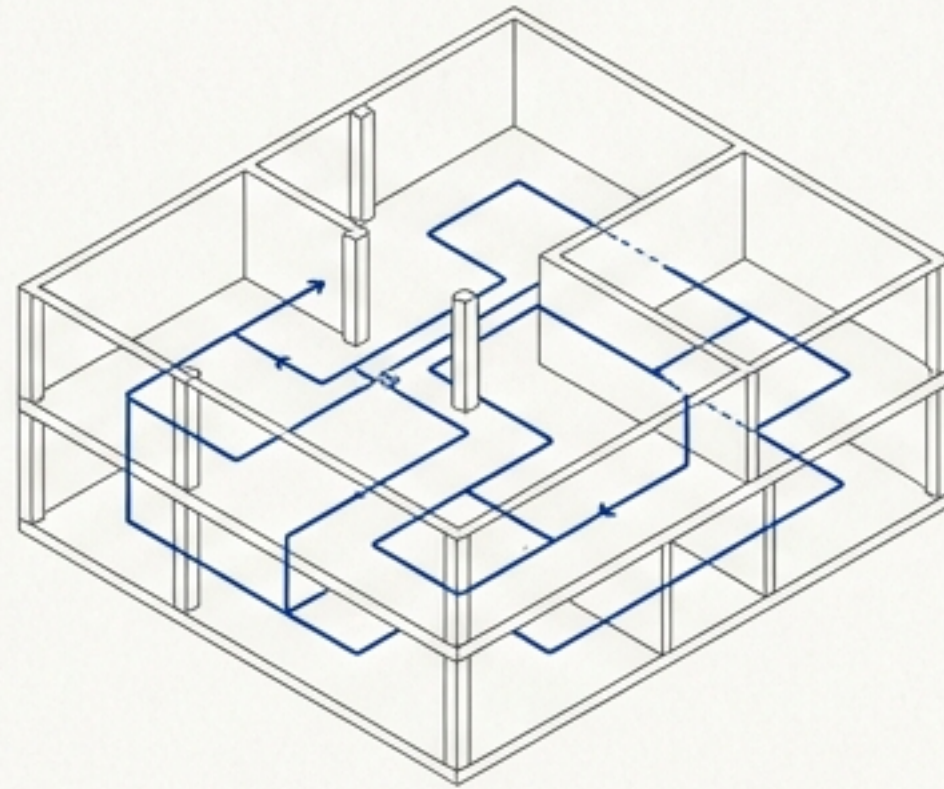
Stage 1 v1.0 - The Foundation



Established core spatial hierarchy, classification, and dedicated space models.

Missing: Process coupling, topology, equipment, systems.

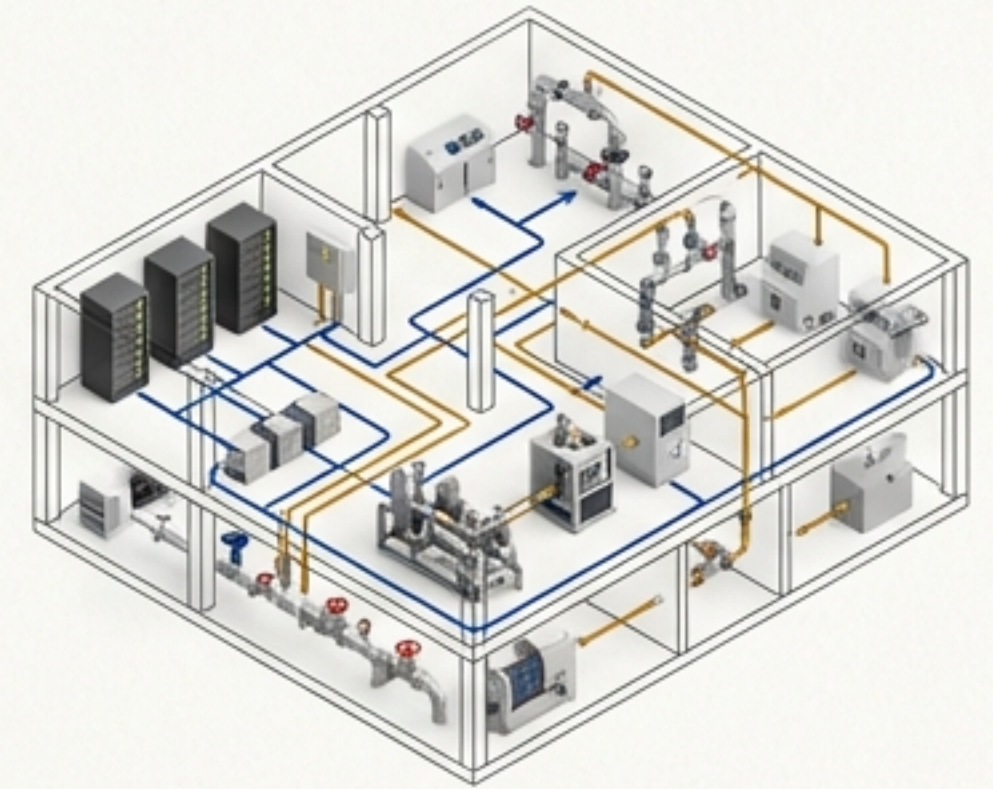
Stage 2 v1.1 - The Connections



+ Integrated process coupling and spatial topology.

Missing: Equipment configuration, system hardware, information systems.

Stage 3 v1.2 - The Living Model



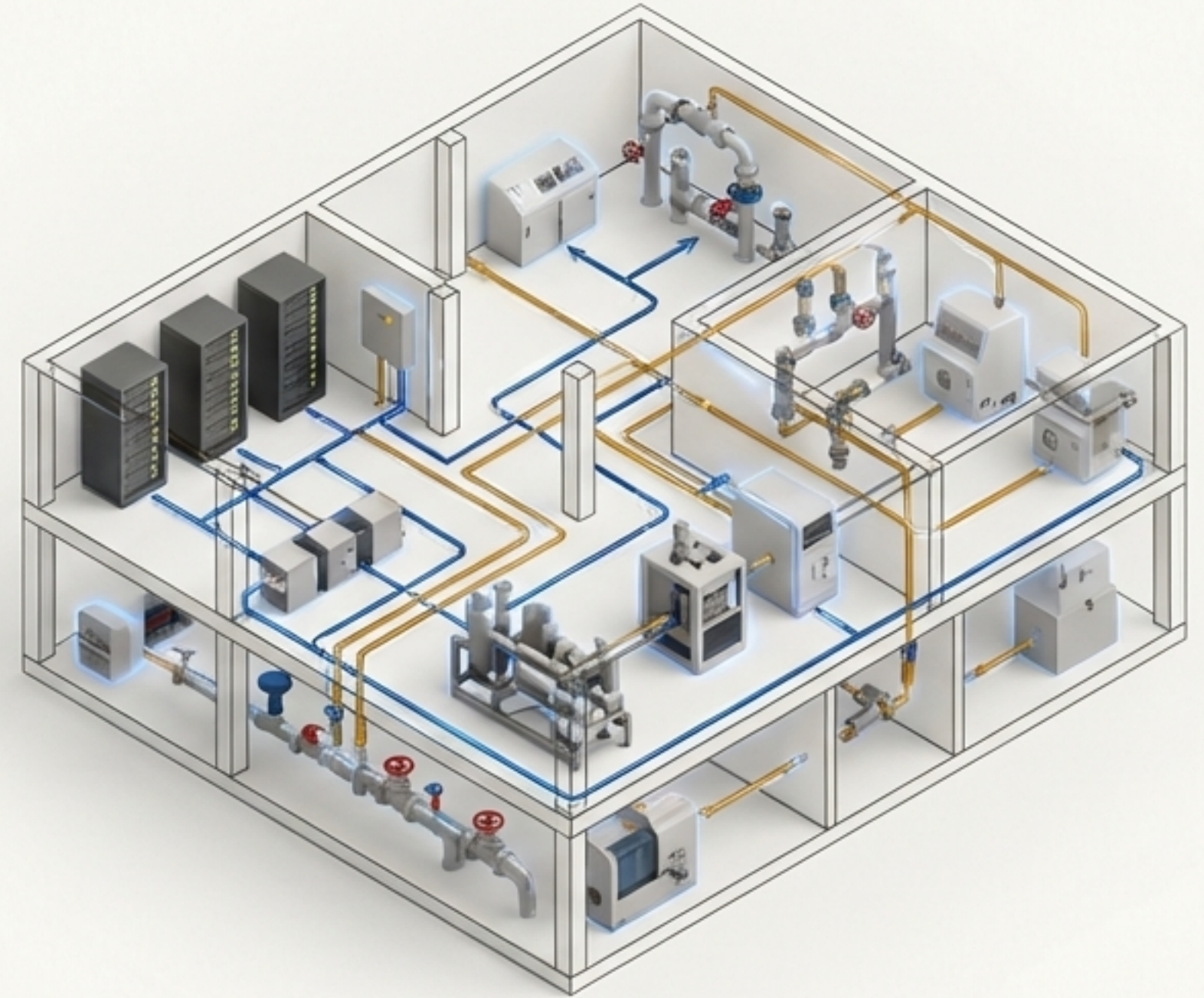
+ Added full equipment configuration, system coupling, O&M, and emergency protocols.

Status: Complete.

v1.2 Achieves a New Threshold of Operational Fidelity

v1.2

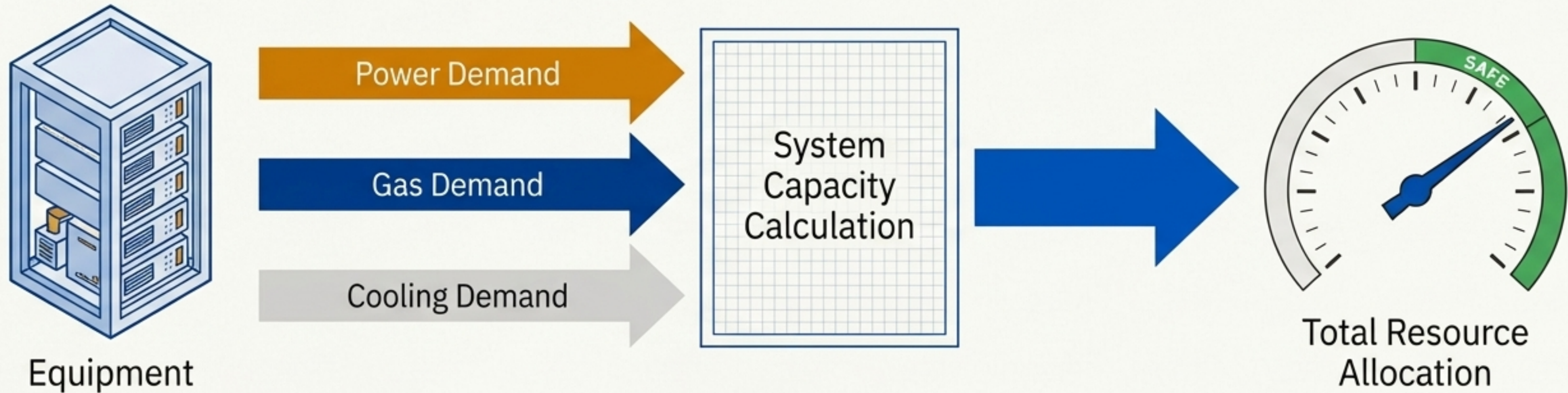
By integrating equipment, systems, operations, and emergency planning, Agent-02 moves beyond static representation. It now functions as a complete digital twin, enabling dynamic analysis operational foresight.



A Closed-Loop Demand Chain Guarantees Resource Allocation

For the first time, there is an unbroken, calculated link from the specific demands of each piece of equipment to the overall capacity of the supporting systems.

Every device's requirement for Power, Gas, and Cooling is now precisely quantified and mapped directly against total system supply, eliminating guesswork.



System Endpoints are Defined with Absolute Spatial Precision

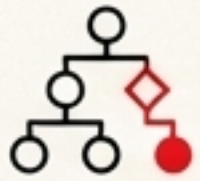
The model moves beyond general topology to define the exact physical coordinates and parameters of every system endpoint.

This includes the precise physical location, height, quantity, and spacing for all terminal connections, providing the ground-truth for installation and operations.



Operations & Maintenance are Integrated by Design, Not as an Afterthought

Agent-02 v1.2 embeds operational intelligence directly into the ontology, allowing us to anticipate and plan for the entire lifecycle of the space from day one.



Pre-Analyzed Failure Modes

Systematically identifying potential points of weakness before they manifest.



Pre-Defined Maintenance Plans

Maintenance schedules and procedures are built directly into the model's components.



Pre-Set Monitoring Configurations

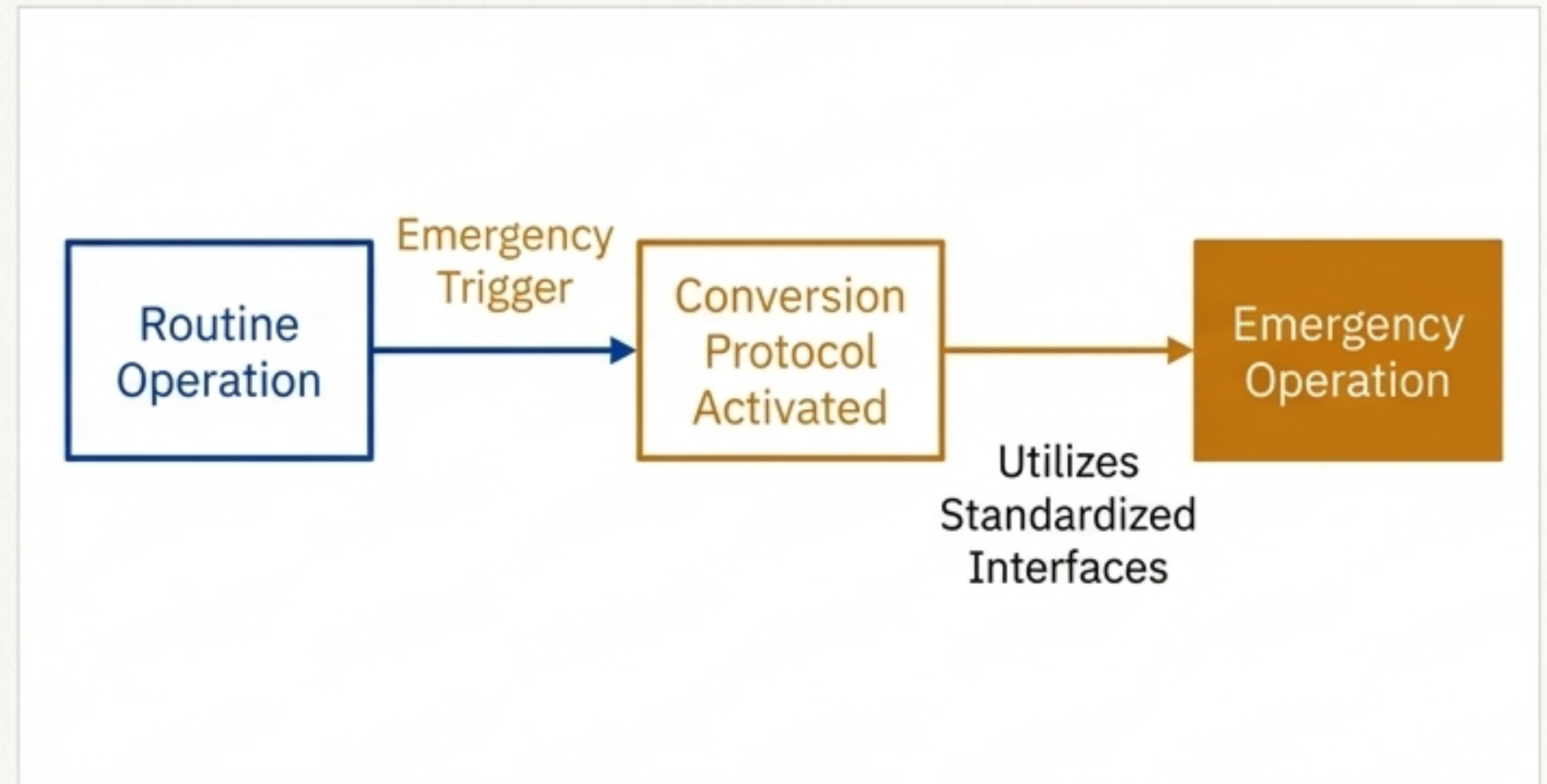
Health-check parameters and alert thresholds are defined upfront for all critical systems.

The Model Systematizes the Shift Between Routine and Emergency States

The ontology is engineered for operational resilience, with a defined and repeatable process for transitioning between normal and crisis scenarios.

Core Features

- **Standardized Interfaces:** Ensures seamless and predictable connection to reserved capacity and backup systems.
- **Proceduralized Conversion Flows:** The exact steps required to switch between operational modes are defined, removing ambiguity under pressure.



Agent-02 v1.2 is the Foundation for the Next Phase of Development

The completion of the v1.2 Space Ontology delivers a robust, high-fidelity digital twin that is ready for immediate use. It fully supports and de-risks the subsequent modeling work for Agent-05 and Agent-08, providing a complete and validated source of truth.

